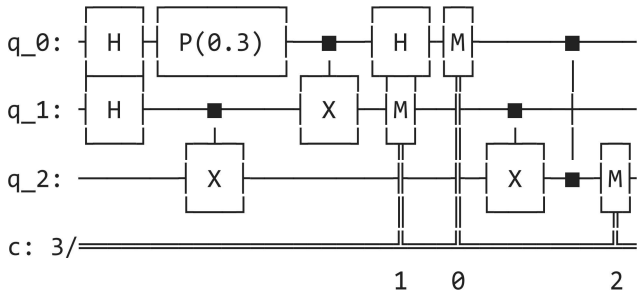
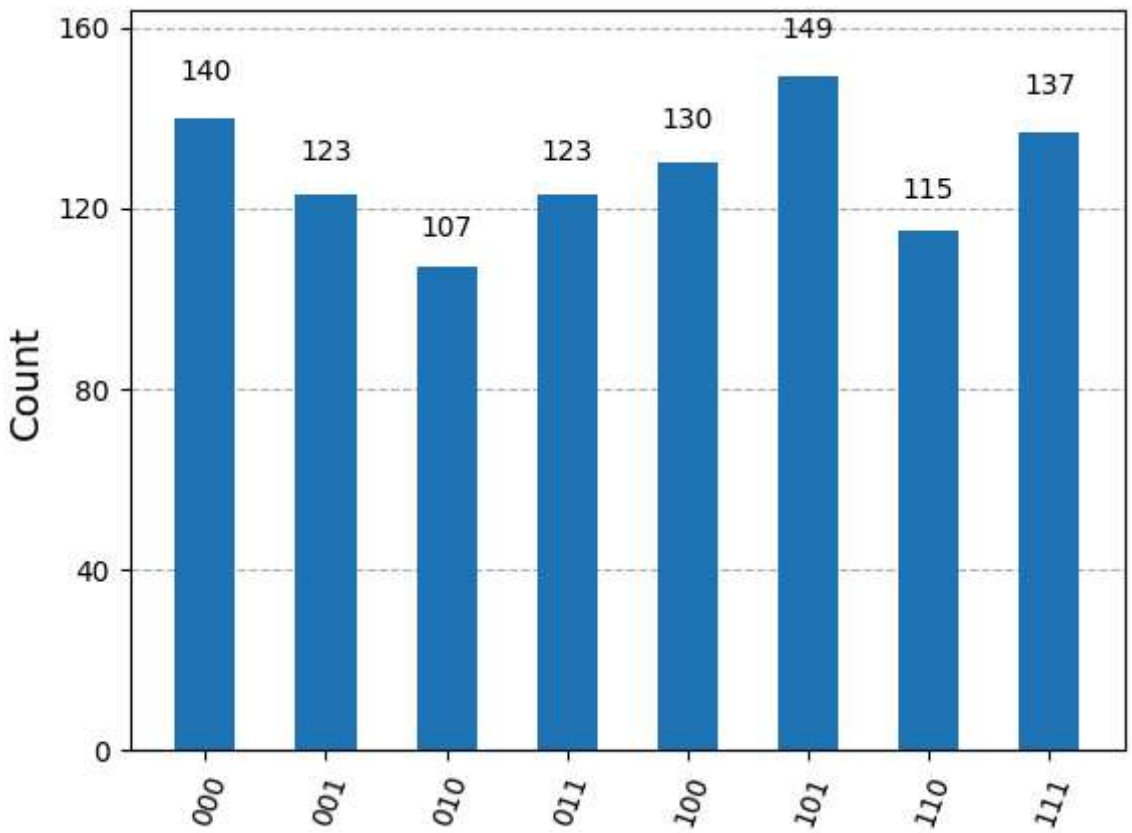
In [1]:
```
#Based on the theory which we have learned for quantum teleportation, qiskit code i
#Quantum Teleportation Protocol
from qiskit import QuantumCircuit
from qiskit_aer import Aer
from qiskit.visualization import plot_histogram
from qiskit.quantum_info import Statevector
import matplotlib.pyplot as plt

# Step 1: Create a 3-qubit circuit
qc = QuantumCircuit(3, 3)

# Step 2: Prepare an arbitrary state to teleport (example: |ψ⟩ = α|0⟩ + β|1⟩)
qc.h(0)          # put qubit0 in |+>
qc.p(0.3, 0)     # add a phase

# Visualize initial state
initial_state = Statevector.from_instruction(qc)
print("State to teleport:", initial_state)

# Step 3: Create entanglement between qubit1 and qubit2
qc.h(1)
qc.cx(1, 2)

# Step 4: Bell measurement on qubit0 & qubit1
qc.cx(0, 1)
qc.h(0)
qc.measure([0, 1], [0, 1])

# Step 5: Conditional corrections for Bob (qubit2)
qc.cx(1, 2)
qc.cz(0, 2)

# Step 6: Measure Bob's qubit
qc.measure(2, 2)

print(qc.draw())

# Simulator backend
sim = Aer.get_backend("aer_simulator")

# Run the circuit
qc_compiled = qc.copy()
qc_compiled = sim.run(qc_compiled, shots=1024).result()

# Get counts
counts = qc_compiled.get_counts()
print("Measurement results:", counts)
plt.show()
# Plot results
plot_histogram(counts)
#plt.show()
```

```
State to teleport: Statevector([0.70710678+0.j        , 0.67552491+0.20896434j,
            0.        +0.j        , 0.        +0.j        ,
            0.        +0.j        , 0.        +0.j        ,
            0.        +0.j        , 0.        +0.j        ],
           dims=(2, 2, 2))
```




```
Measurement results: {'011': 123, '000': 140, '101': 149, '001': 123, '100': 130, '1
11': 137, '110': 115, '010': 107}
```

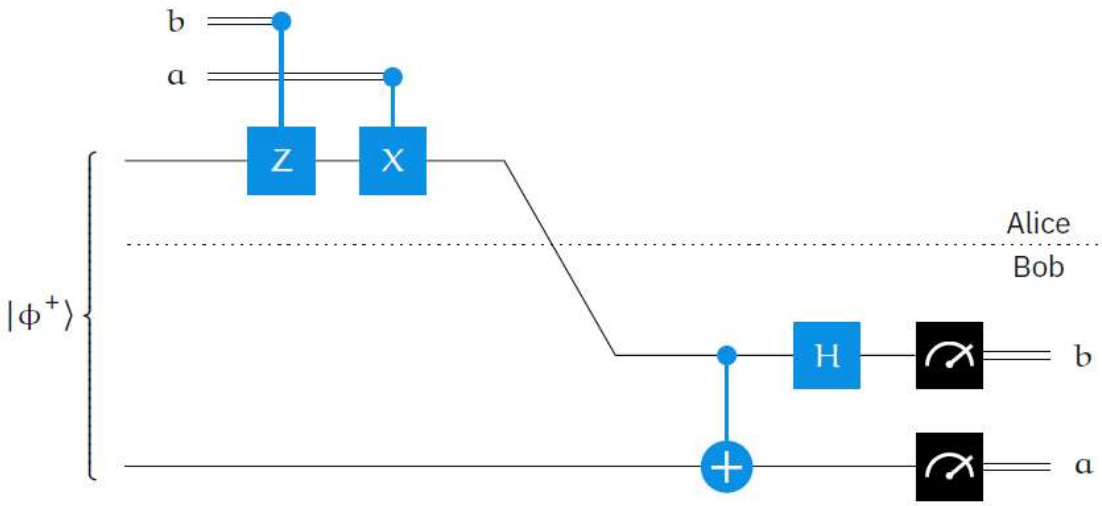
Out[1]:



Task to be submitted on two classical bit transmittion using a single qubit (superdense coding)

Algorithm to build superdense coding

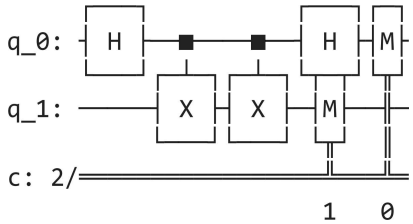
Take the reference of following diagram



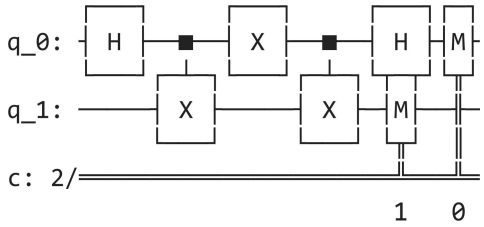
In [3]:
```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import Aer
from qiskit.visualization import plot_histogram
import matplotlib.pyplot as plt

# Simulator
sim = Aer.get_backend('aer_simulator')

messages = {
    '00': (0, 0),
    '01': (1, 0),
    '10': (0, 1),
    '11': (1, 1)
}

results = {}

for msg, (a, b) in messages.items():
    print(f"\nEncoding message {msg}")

    # Step 1: Create circuit with 2 qubits and 2 classical bits
    qc = QuantumCircuit(2, 2)

    # Step 2: Create entangled pair |Φ+>
    qc.h(0)
    qc.cx(0, 1)

    # Step 3: Alice encodes her 2 classical bits
    if b == 1:
        qc.z(0)   # Apply Z if bit b = 1
    if a == 1:
        qc.x(0)   # Apply X if bit a = 1

    # Step 4: Alice sends her qubit to Bob
    # (No code step needed — just a conceptual transfer)

    # Step 5: Bob decodes
```

```
    qc.cx(0, 1)
    qc.h(0)

    # Step 6: Bob measures both qubits
    qc.measure([0, 1], [0, 1])

    # Draw the circuit
    print(qc.draw())

    # Step 7: Run on simulator
    compiled = transpile(qc, sim)
    result = sim.run(compiled, shots=1024).result()

    # Get results
    counts = result.get_counts()
    print("Decoded bits:", counts)
    results[msg] = counts

plt.show()
plot_histogram(list(results.values()), legend=list(results.keys()))
```

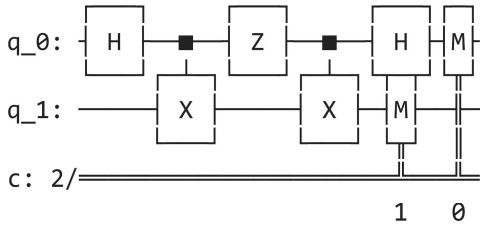
```
Encoding message 00
```



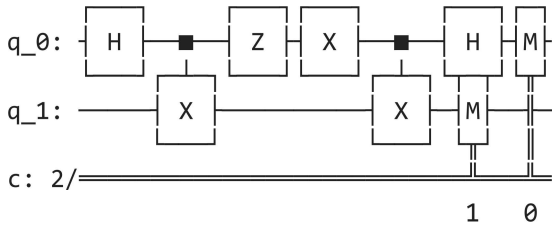

```
Decoded bits: {'00': 1024}

Encoding message 01
```




```
Decoded bits: {'10': 1024}

Encoding message 10
```

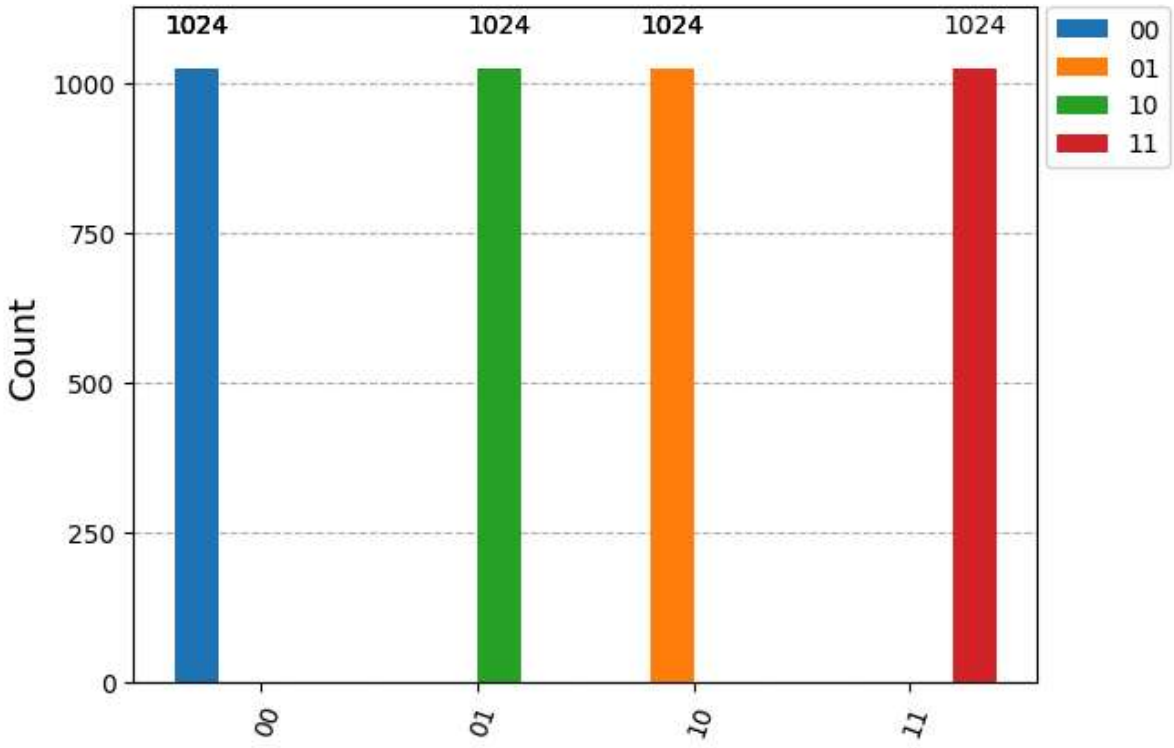



```
Decoded bits: {'01': 1024}

Encoding message 11
```




```
Decoded bits: {'11': 1024}
```

Out[3]:


1024
1024
1024
1024
1000
750
500
250
0
Count
00
01
10
11
00
01
10
11